

PAVAN D R

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SUMMARY

Final-year Computer Science Engineering student with strong foundations in Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, and Software Development. Proficient in Python, Java, SQL, React, FastAPI, and machine learning, with hands-on experience developing full-stack, AI/ML, and database-driven applications. Interested in building secure, scalable, and user-focused intelligent systems.

PROJECTS

Full-Stack Retrieval-Augmented Generation (RAG) Application

Technologies: React, Vite, TypeScript, Tailwind CSS, FastAPI, ChromaDB, Sentence Transformers, Gemini API, SQLite, JWT

- Developed a full-stack RAG application enabling users to upload PDF, DOCX, and TXT documents, perform semantic retrieval, and receive AI-generated answers grounded in their documents.
- Implemented the RAG pipeline using document extraction, semantic chunking, **all-MiniLM-L6-v2** embeddings, ChromaDB vector retrieval, and Gemini-based response generation with source citations.
- Built a layered FastAPI backend using clean architecture principles across API, application, domain, and infrastructure layers, with persistent chat sessions, conversation management, document indexing, and streaming chat support.
- Implemented JWT access/refresh authentication, bcrypt password hashing, password reset, protected routes, role support, and user-isolated document and conversation data.
- Designed a Claude-style React interface with document management, chat history, source citation cards, confidence indicators, summaries, key points, and follow-up questions.

CricShift – Momentum Shift Detection and Win Prediction in Cricket

Technologies: Python, Machine Learning, Pandas, NumPy, Scikit-learn, SQL, Data Visualization, Flask

- Developed an ML-based system to detect momentum shifts and predict win probability in cricket matches using ball-by-ball data.
- Performed data preprocessing and feature engineering using match, player, venue, and performance-related features.
- Built predictive models and visualizations to analyze momentum changes and identify factors influencing match outcomes.

FurEver Pet Care Web Application

Technologies: Python, Flask, HTML, CSS, JavaScript, SQLite

- Developed a database-driven pet care web application covering pet services, grooming, boarding, products, and care-related functionality.
- Implemented product browsing, service discovery, category-based filtering, and booking functionality with a responsive user interface.
- Integrated Flask with SQLite to manage products, services, bookings, and pet-related data.

SKILLS & ACCOMPLISHMENTS

Programming Languages: Python, Java, C, SQL

Web & Backend: HTML, CSS, JavaScript, React, TypeScript, FastAPI, Flask, REST APIs

AI/ML & Data: Machine Learning, RAG, Pandas, NumPy, Scikit-learn, Sentence Transformers, Data Preprocessing, Feature Engineering, Data Visualization

Databases & Tools: SQLite, ChromaDB, Git, GitHub, VS Code, Jupyter Notebook

Core Computer Science: Data Structures and Algorithms, OOP, Operating Systems, DBMS

EDUCATION

K S School of Engineering and Management

2023 – Present

Bachelor of Engineering (B.E.) – Computer Science Engineering

Bengaluru, Karnataka

CGPA: 8.56

CERTIFICATIONS

- AWS Academy Graduate – Cloud Developing — AWS Academy — 2026
- AWS Academy Graduate – Cloud Foundations — AWS Academy — 2026
- Explore Machine Learning using Python — Infosys Springboard — 2026
- Getting Started with Artificial Intelligence — IBM SkillsBuild — 2024

LANGUAGES

English — Kannada — Hindi — Telugu